

📖 Master Architectural Mindmap



Implementation Verification Matrix

The following table correlates every mindmap branch with the verified status in the codebase:

Mindmap Branch	Specific Component	Codebase Implementation Reference	Status
Mobile	Executive Dashboard	<code>lib/features/dashboard/presentation/dashboard_screen.dart</code>	 IMPLEMENTED
Mobile	Monitored Repositories	<code>lib/features/repositories/presentation/repositories_screen.dart</code>	 IMPLEMENTED
Mobile	SOC Alert Triage	<code>lib/features/alerts/presentation/alerts_screen.dart</code>	 IMPLEMENTED
Mobile	AI Security Copilot	<code>lib/features/ai/presentation/ai_assistant_screen.dart</code>	 IMPLEMENTED
Mobile	Compliance PDF Generator	<code>lib/core/services/report_pdf_service.dart</code>	 IMPLEMENTED
Mobile	Settings & Diagnostics	<code>lib/features/settings/presentation/settings_screen.dart</code>	 IMPLEMENTED
Cloud	FastAPI Asynchronous Server	Cloud Microservice (<code>secureguard-backend-7eqm.onrender.com</code>)	 IMPLEMENTED
Cloud	REST Endpoints Contract	<code>lib/core/network/api_endpoints.dart</code>	 IMPLEMENTED
Cloud	WebSocket Threat Engine	<code>lib/core/network/websocket_service.dart</code>	 IMPLEMENTED
Cloud	Database Persistence	Cloud PostgreSQL + <code>lib/core/storage/hive_storage_service.dart</code>	 IMPLEMENTED
Sources	Wazuh Syslog Ingestion	<code>lib/features/alerts/data/alerts_repository.dart</code>	 IMPLEMENTED
Sources	Wazuh Manager REST API	Direct remote daemon control	 PLANNED
Sources	GitHub Telemetry & Webhooks	<code>lib/features/repositories/data/repository_repository.dart</code>	 IMPLEMENTED
Sources	GitHub OAuth2 Deep-Links	Interactive in-app OAuth2 callback redirect	 PLANNED
Sources	Semgrep SAST Scanning	<code>lib/repositories/scan_repository.dart</code>	 IMPLEMENTED
Sources	Custom In-App Rule Editor	Mobile YAML rule authoring editor	 PLANNED

Mindmap Branch	Specific Component	Codebase Implementation Reference	Status
Processing	Syslog Event Normalization	<code>lib/features/alerts/models/alert_model.dart</code>	 IMPLEMENTED
Processing	Alert Lifecycle State Machine	<code>updateAlertStatus()</code> in <code>AlertsRepository</code>	 IMPLEMENTED
AI	Heuristic CVE Remediation	<code>AiRepositoryImpl._generateMockResponse()</code>	 IMPLEMENTED
AI	Cloud LLM Chat Bridge	<code>POST /v1/ai/chat</code> client dispatch	 IMPLEMENTED
AI	Autonomous PR Remediation	One-tap GitHub Pull Request patch generation	 PLANNED
AI	On-Device Quantized SLM	Gemma-2B / Llama-3-1B via ONNX Runtime	 PLANNED
Security	Hardware Biometric Gate	<code>lib/core/services/biometric_service.dart</code> (<code>local_auth</code>)	 IMPLEMENTED
Security	Hardware Keystore Encryption	<code>lib/core/storage/secure_storage_service.dart</code> (AES-256)	 IMPLEMENTED
Security	SHA256 PDF Audit Stamping	<code>ReportPdfService._generateAuditSignature()</code>	 IMPLEMENTED
Security	Multi-Tenancy per Org	Dynamic organization switching & scoped tokens	 PLANNED
Security	Per-Device Mutual TLS (mTLS)	Client X.509 certificate enrollment	 PLANNED